

The relative dimension of the four-dimensional Boolean lattice

Xiaojian Zeng

PBC School of Finance, Tsinghua University, 100190

xiaojianzeng@foxmail.com

Abstract

For the Boolean lattice $\mathcal{B}_4$, we determine the minimum total length of a local realizer to be 42, and hence its relative dimension to be $21/8$. Four explicit partial linear extensions attain this value. An exact rational covering certificate gives the fractional optimum $202/5$; excluding integral length 41 requires a further argument. Complete enumeration of partial extensions with small dual slack forces a hypothetical length-41 realizer to have a full extension and at most five nonempty members. A Boolean encoding of these families has a separately checked DRAT refutation. We give the reduction, including the full-extension anchor check, and describe the exact certificates and their reproduction. This is a finite computational result; the asymptotic behavior of relative dimension for Boolean lattices is not settled here.

1 Definition, scope, and the attaining family

A partial linear extension (PLE) of a finite poset P is a linear extension of the order induced on an arbitrary subset of P . Its length is the size of that subset. Following Dürschnabel, Hodor, Micek, Stumme, and Trotter [1], a family $\mathcal{L}$ of PLEs is a local realizer if every two elements co-occur in some member and

$$x \leq_P y \iff x \leq_L y \text{ for every } L \in \mathcal{L} \text{ containing } x, y.$$

For nonempty P its relative dimension is

$$\text{rdim}(P) = \min_{\mathcal{L}} \frac{\sum_{L \in \mathcal{L}} |L|}{|P|}.$$

In particular, the objective is average occurrence, rather than the maximum occurrence used in local dimension.

Let $\mathcal{B}_4$ consist of all subsets of $[4] = \{1, 2, 3, 4\}$, ordered by inclusion. Encode a subset A as the integer $\sum_{i \in A} 2^{i-1}$, so that the minimum and maximum are 0 and 15. We use masks and their subsets interchangeably. A PLE is a sequence of distinct masks with no later element properly contained in an earlier one. Define the directed requirement set

$$U = \{(a, b) : b \not\subseteq a\}. \tag{1}$$

It has $65 = 3^4 - 2^4$ strict comparable rows and 110 directed incomparable rows, hence 175 rows altogether. A family is a local realizer exactly when every row $(a, b) \in U$ occurs with a before b in a member: comparable pairs require co-occurrence, and incomparable pairs require both orders. Each element participates in a distinct-pair requirement, so these conditions also imply occurrence for the diagonal case $x = y$.

Theorem 1. *The minimum total length of a local realizer of $\mathcal{B}_4$ is 42. Thus $\text{rdim}(\mathcal{B}_4) = 21/8$. The fractional relaxation of the covering problem on U , with cost $|L|$ for a PLE L , has optimum $202/5$.*

The following four PLEs have total length $2 + 10 + 14 + 16 = 42$:

$$\begin{aligned} L_1 &= (14, 1), \\ L_2 &= (12, 1, 9, 5, 13, 2, 10, 6, 14, 3), \\ L_3 &= (8, 2, 10, 1, 9, 3, 11, 4, 12, 6, 14, 5, 13, 7), \\ L_4 &= (0, 4, 2, 6, 1, 5, 3, 7, 8, 12, 10, 14, 9, 13, 11, 15). \end{aligned} \tag{2}$$

Testing all forward pairs in these sequences verifies the PLE condition and coverage of all 175 rows, proving the upper bound. The remaining sections prove the lower bound by exact finite computation.

The source paper [1, Lemma 4 and the proof of Theorem 5] gives product subadditivity and the upper bounds $1, 3/2, 2$ for $\mathcal{B}_1, \mathcal{B}_2, \mathcal{B}_3$. In particular it gives $\text{rdim}(\mathcal{B}_4) \leq 3$. The contribution here is the exact finite value, an attaining family, and a universal integral lower certificate; the fractional bound alone would allow total length 41. The asymptotic question in [1, Section 4, Question 2] remains outside this result. Our comparison concerns the indicated statements and proofs of version 1; we make no comprehensive priority or latest-literature claim. As adjacent computational methodology, Zeng [2] also uses certified Boolean encodings and checked refutations in finite combinatorics. That reference is not used to establish any assertion about relative dimension.

2 A rational certificate and complete slack enumeration

Write A for the atoms (one-element subsets) and C for the coatoms (three-element subsets). Give the rows of U the nonnegative integer weights in Table 1; every unlisted weight is zero.

Table 1: The weight function w on directed requirements.

Row (a, b)	Count	$w(a, b)$
0 to an atom, or a coatom to 15	8	5
Distinct atoms, or distinct coatoms	24	8
A two-set to a disjoint atom, or a coatom to a two-set whose union with it is [4]	24	20
A coatom to its complementary atom	4	24

The total weight is $8 \cdot 5 + 24 \cdot 8 + 24 \cdot 20 + 4 \cdot 24 = 808$. For a PLE L define its scaled dual slack by

$$r(L) = 20|L| - \sum_{a \text{ before } b \text{ in } L} w(a, b). \tag{3}$$

Here weights outside U are understood to be zero.

Lemma 2 (Exact finite certificate). *Every PLE of $\mathcal{B}_4$ has $r(L) \geq 0$. There are 85 140 nonempty PLEs with $r(L) \leq 12$, on 11 060 domains, with the following histogram:*

$r(L)$	0	4	8	9	10	12
number	22 500	18 000	13 200	18 000	9 000	4 440.

For every such PLE the following properties hold:

1. It reverses at most one of the four critical pairs $(15 - a, a)$, $a \in \{1, 2, 4, 8\}$. If it reverses none, its slack is 12.
2. If its slack is neither 9 nor 10, inclusion of 0 forces inclusion of all atoms, and inclusion of 15 forces inclusion of all coatoms.
3. Among these PLEs, exactly 576 have full domain. Every one of these full extensions has slack 0 and reverses exactly one critical pair.

Computation and completeness argument. For a domain $S \subseteq \mathcal{B}_4$ let $D(S)$ be the largest value of $-r(L)$ over its linear extensions. Deleting the last element gives the exact recurrence

$$D(\emptyset) = 0, \quad D(S) = \max_{x \text{ maximal in } S} \left(D(S \setminus \{x\}) + \sum_{a \in S \setminus \{x\}} w(a, x) - 20 \right). \quad (4)$$

Conversely, an extension of $S \setminus \{x\}$ followed by a maximal element x is an extension of S . Induction therefore proves the recurrence for all $2^{16} = 65\,536$ domains, without any restriction to order ideals. Exact integer evaluation gives $\max_S D(S) = 0$.

Reconstructing paths in this recurrence enumerates all extensions with score $-r(L) \geq -12$. At a state (S, h) , a branch with last element x is omitted only if the corresponding expression in (4) is less than h ; otherwise the same reconstruction is applied to the predecessor with the threshold decreased by the last-element increment. The exact maximum makes this pruning complete. Distinct last-element paths give distinct orders. Direct inspection of the resulting sequences gives the histogram and the three stated properties, including the full-domain test in item 3.

A separate first-element recurrence deletes a minimal element:

$$D(S) = \max_{x \text{ minimal in } S} \left(D(S \setminus \{x\}) + \sum_{b \in S \setminus \{x\}} w(x, b) - 20 \right).$$

Its reconstruction gives exactly the same set of 85 140 sequences. An additional unpruned enumeration starts with the empty sequence and appends each unused x for which the prefix contains no strict superset of x . These are precisely the permissible next elements. Induction on the prefix length shows that it visits every nonempty PLE exactly once. It visits 129 298 051 PLEs and confirms nonnegative slack, the histogram, and items 1–2. Item 3 is checked on the full-domain entries of the complete catalogue by the first-element verifier.

For clarity, the full-extension anchor in item 3 is a separate checked assertion, not an inference from items 1–2. A further first-element dynamic program that forbids all four critical reversals gives minimum slack 24 for a full extension with no reversal. This exceeds the threshold 12. The scripts, exact outputs, and reproduction instructions are identified in Section 5.

The at-most-one part of item 1 also follows without enumeration. Reversing two distinct complementary pairs (c_i, a_i) and (c_j, a_j) would give $c_i <_L a_i <_L c_j <_L a_j <_L c_i$, because $a_i \subset c_j$ and $a_j \subset c_i$, which is impossible in a linear order. $\square$

Nonnegative slack means that $w/20$ is a feasible dual solution for the fractional covering problem. Summing weighted coverage yields

$$T := \sum_{L \in \mathcal{L}} |L| \geq \frac{808}{20} = \frac{202}{5}. \quad (5)$$

Consequently an integral realizer has $T \geq 41$.

To specify a matching fractional cover, use the four representatives and total masses

representative	L_2	L_3	L_4	$(5, 2, 7, 8, 11, 12)$
total mass	$2/5$	$3/5$	1	2.

For each representative apply the 24 coordinate permutations, either alone or followed by complementation and reversal of the sequence. Give each of these 48 operations $1/48$ of its specified mass, adding the masses of duplicate images. Each image remains a PLE. Exact rational summation gives 96 distinct positive columns and coverage at least 1 on every row of U , with minimum coverage exactly 1. Their cost is

$$10(2/5) + 14(3/5) + 16 + 6 \cdot 2 = 202/5.$$

Together with (5), this proves the fractional assertion of Theorem 1. Symmetry averaging is used only to describe this fractional certificate; no orbit condition is imposed on integral realizers.

3 The full-extension reduction at total length 41

Suppose that a local realizer has total length $T = 41$, and delete empty members. Let $k(a, b)$ be the coverage multiplicity of row (a, b) and let $R = \sum_L r(L)$. Equations (3) and (5) give

$$R + \sum_{(a,b) \in U} w(a, b)(k(a, b) - 1) = 20 \cdot 41 - 808 = 12. \quad (6)$$

Every term is a nonnegative integer. Thus each chosen PLE lies in the catalogue of Lemma 2, and all weight-20 and weight-24 rows occur exactly once.

Let u be the sum of excess multiplicities on weight-8 rows and e the corresponding sum on weight-5 rows. Then

$$R + 8u + 5e = 12. \quad (7)$$

All sums of allowed slacks that do not exceed 12 lie in $\{0, 4, 8, 9, 10, 12\}$. Checking these six possibilities in (7) gives exactly

$$(R, u, e) = (4, 1, 0) \quad \text{or} \quad (12, 0, 0).$$

A total slack of 4 cannot contain a summand 9 or 10; at total slack 12, the remaining slack would be 3 or 2, smaller than the smallest positive slack 4. Also, $e = 0$ says that every weight-5 row occurs exactly once.

By Lemma 2(2), every occurrence of 0 includes all atoms, and hence covers all four weight-5 rows from 0. Therefore 0 occurs in exactly one PLE. The same argument puts 15 in exactly one PLE. The row $(0, 15)$ places them in the same member F . Every row $(0, x)$ with $x \neq 0$ must then be covered by F , so F has full domain.

Each of the four critical pairs is a weight-24 row and occurs exactly once. By Lemma 2(1), their four PLEs are distinct. Any further nonempty PLE has no critical reversal and hence slack 12. The budget (6) allows at most one such further member. There are thus at most five nonempty PLEs.

The anchor check in Lemma 2(3) now applies to F : it reverses exactly one critical pair. Relabel coordinates so that this pair is $(14, 1)$ and call this full member slot 0. Label the other three critical slots by the atoms 2, 4, 8. Permuting those three coordinates makes their order in F equal to $2 <_F 4 <_F 8$. Use slot 4 for any extra PLE, with empty padding if absent. These operations are poset automorphisms and permutations of family members, so they preserve every integral solution.

4 Boolean encoding and checked exclusion

We encode the normalized five slots by a conjunctive normal form (CNF). For $i \in \{0, 1, 2, 3, 4\}$ and masks a , let $p_{i,a}$ mean that a is present in slot i . For numerical $a < b$, let $o_{i,a,b}$ mean that a precedes b in an auxiliary full order. Comparable pairs have their forced poset orientation. For each numerical triple $a < b < c$, include

$$(\neg o_{i,a,b} \vee \neg o_{i,b,c} \vee o_{i,a,c}), \quad (o_{i,a,b} \vee o_{i,b,c} \vee \neg o_{i,a,c}).$$

These exclude the two directed triangles. A tournament with no directed triangle is transitive: a shortest directed cycle of length at least four has a chord producing a shorter cycle. Thus the auxiliary orders are full linear extensions. Their restrictions to the present elements are PLEs.

Conversely, every PLE extends to a full linear extension. Add its forward order relations to the poset. A cycle in the augmented relation would have to use an added relation. Between successive added relations, each poset path whose endpoints are PLE elements goes forward in their given order; each added relation also goes forward. Traversing a cycle would strictly increase positions in the PLE and return to the initial position. This is impossible. A topological ordering of the augmented relation supplies the required full extension. Consequently the auxiliary orders lose no PLEs.

For $(a, b) \in U$ let $t_i(a, b)$ denote the orientation literal for a before b (either $o_{i,a,b}$ or $\neg o_{i,b,a}$). Four clauses encode

$$c_{i,a,b} \iff p_{i,a} \wedge p_{i,b} \wedge t_i(a, b) :$$

three implications from $c_{i,a,b}$ to its factors, and the reverse clause $\neg p_{i,a} \vee \neg p_{i,b} \vee \neg t_i(a, b) \vee c_{i,a,b}$. For each of the 175 rows include $\bigvee_{i=0}^4 c_{i,a,b}$. For weight-20 and weight-24 rows include pairwise at-most-one clauses on their five coverage variables, as justified by (6).

The critical coverage variables for $(14, 1), (13, 2), (11, 4), (7, 8)$ are true in slots 0, 1, 2, 3, respectively. All $p_{0,a}$ are true, while $p_{i,0}$ and $p_{i,15}$ are false for $i > 0$. Finally $o_{0,2,4}$ and $o_{0,4,8}$ are true. Section 3 justifies every one of these normalization clauses.

To impose the length bound, list the 80 presence variables as $x_1, \dots, x_{80}$. Introduce $Q(i, j)$ for $1 \leq j \leq \min(i, 42)$, meaning that at least j of the first i variables are true. With $Q(i, 0)$ true and $Q(i, j)$ false when $j > i$, use

$$Q(i, j) \iff Q(i-1, j) \vee (x_i \wedge Q(i-1, j-1)). \quad (8)$$

Each gate $v \iff A \vee (x \wedge B)$ has the four clauses

$$(\neg A \vee v), \quad (\neg x \vee \neg B \vee v), \quad (\neg v \vee A \vee x), \quad (\neg v \vee A \vee B).$$

Induction on i proves the stated counting semantics, including the truncation at 42. The final unit clause $\neg Q(80, 42)$ imposes total length at most 41.

The resulting CNF, `evidence/length41-full.cnf`, has 4055 variables and 19908 clauses. A hypothetical length-41 realizer induces a satisfying assignment by the preceding reduction. CaDiCaL 3.0.1 generated the binary DRAT certificate `evidence/length41-full.drat`. The separate `drat-trim` checker verified it against this exact CNF, returning `s VERIFIED`. The check reports 70 633 core lemmas and 3 983 156 resolution steps, with zero RAT lemmas in the core. Thus it certifies a propositional refutation of the formula, not merely a solver’s reported optimum.

An independent semantic reconstruction gives the same clause multiset as the stored CNF. As a boundary test, the length-42 family (2), after normalization and completion of its auxiliary orders, satisfies every clause except $\neg Q(80, 42)$. This test checks consistency of the encoding; the DRAT refutation supplies its unsatisfiability proof. Therefore $T = 41$ is impossible. Equation (5) excludes $T \leq 40$, and (2) attains 42, completing Theorem 1.

5 Reproduction and limitations

The accompanying computational files form part of the proof. The principal artifacts are the following; all paths are relative to the project root.

- `evidence/replay_lp.py` checks the upper family, the first-element dual recurrence, and the rational fractional cover using the Python standard library. The data are in `evidence/lp-certificate.json`.
- `evidence/near_tight_columns.py` reconstructs the complete catalogue `evidence/near-tight-columns.json`; `evidence/exhaustive_slack.cpp` provides the independent unpruned enumeration.
- `evidence/encode_length41.py` and `evidence/extreme_reduction.py` construct the normalized CNF. The two `length41-full` files identified in Section 4 are the formula and its checked refutation.
- `audit/referee_verify.py` independently reconstructs the weights, first-element catalogue, full-extension anchor, fractional cover, and CNF semantics. Its exact results are in `audit/referee-verification.json`.

Reproduction should use a disposable copy of the supplied project because the Python programs regenerate some output files. From that copy’s root, a compact verification sequence is

```
python3 audit/referee_verify.py
clang++ -O3 -std=c++17 evidence/exhaustive_slack.cpp -o verify-all
./verify-all
drat-trim evidence/length41-full.cnf evidence/length41-full.drat
```

Compile without `-DNDEBUG`, so that the C++ assertions remain active. The independent verifier uses exact integer and rational arithmetic and does not import the discovery implementations. The unpruned enumeration and DRAT check were replayed separately. Discovery also used floating-point optimization, but no floating-point value serves as a proof certificate.

The mathematical scope is $\mathcal{B}_4$ with all 16 elements and all 175 directed requirements. We do not classify all attaining realizers or determine exact values for larger Boolean lattices. The computation is not a formalization in a proof assistant; it retains the usual trust in implementations, the compiler, and the proof checker. Separate recurrence implementations, exact catalogue comparison, semantic CNF reconstruction, and certificate replay provide checks of distinct parts of that trust chain.

Computational and AI assistance. OpenAI Codex assisted with the mathematical search, program construction, certificate analysis, and manuscript preparation. A separate model context performed the mathematical audit and independent reconstruction described above. Such a review is neither human peer review nor evidence of statistical independence. The exact finite certificates, together with the completeness and encoding arguments, are the basis of the computational proof.

References

- [1] D. Dürrschnabel, J. Hodor, P. Micek, G. Stumme, and W. T. Trotter, *Relative Dimension of Posets*, arXiv:2609.05166v1 (2026). <https://arxiv.org/abs/2609.05166v1>.
- [2] Zijian Zeng, *One-Sided Parity Compression for Certified Sat Attacks on the 33-Point Erdős-Szekeres Problem* (2026). <https://doi.org/10.2139/ssrn.7334758>.